MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY

INFORMATION ON RESEARCH RESULTS

1. General information:

Project code: THS2026-68

Project title: Development of an Edge AI-based System for Multimodal Analysis and Clustering of Flood Rescue Events

Project coordinator: Lê Thị Ngọc Ảnh

Implementing institution: Trường Công nghệ Thông tin và Truyền thông, Đại học Cần Thơ

Duration: May–October 2026 (six months).

2. Objective(s):

Develop an AI-based mobile system to receive and analyze flood rescue information and support decision-making under unstable network conditions, with the aim of reducing loss of life and property damage.

3. Creativeness and innovativeness:

The project proposes an integrated approach combining multimodal data analysis, Edge AI and intelligent rescue-event clustering within a unified system.

Instead of relying entirely on centralized server processing, the proposed system aims to perform AI inference directly on mobile devices using optimized lightweight deep learning models, thereby reducing computational requirements and the transmission of large data files.

Furthermore, the integration of location, time and rescue information for identifying and grouping related events is intended to reduce duplicate reports and support visualization of rescue situations on a digital map.

4. Research results:

For image classification, the model was evaluated on a test set of 256 images. Using seed 1024, the model achieved 75.78% accuracy and 74.26% macro-F1, with three criti-

cal errors. The structured pruning experiment achieved 80.08% accuracy, with 4 critical errors.

Additional ONNX CPU screening of the pruned model used the same 256 images and eight synthetic corruptions. Macro-F1 decreased from 78,26% on clean images to 68,04% at 0.4 brightness. This does not establish robustness on Android or field camera images; inherited corruption labels have not been re-adjudicated.

The app sends image labels and the LR score for storage; the adapter uses urgency or description keywords for clustering/ranking. Five-flag LR fitted to synthetic OR labels is not free-text NLP. The English CrisisMMD informativeness pilot reaches text/fusion macro-F1 of 76,60%/78,35%; effect CIs include zero.

A separate pilot uses AI-translated Vietnamese text and AI labels: 179 training, 58 development and 59 test pairs. Text classifiers, DistilBERT fine-tuning and frozen feature fusion were run offline; frozen fusion reaches 64,91% macro-F1 and 33,33% request recall. Labels derive from source text, not independent human gold. Native-Vietnamese evaluation and application fusion remain unfulfilled; information extraction and camera/Android evaluation remain proposed.

In weak-network simulations with bandwidth of 50 kbit/s and a round-trip time of 600 ms, both metadata-only and compressed-image modes succeeded in 257/257 attempts (100%), compared with 230/257 (89.49%) for original images. The two modes transmitting less data achieved higher transmission success rates than original images under the tested conditions. Each mode was measured separately on a computer; transmission over real mobile networks requires further validation.

RQ1 does not establish Product superiority over independently selected Additive (ARI 0.9072 versus 0.9191; Holm p=0.2560). RQ2 supports exact-copy invariance under fixed grouping, without establishing general alignment gains (revised NDCG@5 0.6669 versus random 0.6763). Across 40 seed-level pairs, revised priority performs worse than nearest-first on simulated harm and deadline misses. Three raw Spearman p-values and five related Holm values remain unreconciled in RQ2 replay.

The available results demonstrate the feasibility of selected prototype components in a research environment. They do not yet establish improvements in real-world rescue coordination or disaster response effectiveness.

5. Products:

The project registers three main scientific and technological products:

– A mobile application supporting disaster rescue, integrating AI models and rescue

request submission functions.

– A web-based rescue information management system featuring a digital map for visualizing rescue locations and clustering related events.

– A trained and optimized multimodal AI model package for analyzing rescue information and supporting deployment on mobile devices.

The model package is registered; the image model, structured LR and separate CrisisMMD pilot do not establish completion of Vietnamese NLP and rescue fusion in the application.

In addition, the project registers a research bulletin, a research summary and a demonstration video of no more than two minutes, in accordance with Can Tho University's requirements. The paper "Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study" has been published in the ISDS 2026 proceedings, Part II, in Communications in Computer and Information Science (CCIS), volume 3253, Springer Nature, 2026.

6. Effects, technology transfer means and applicability:

The outputs support education and research in edge image inference and rescue information management, with planned transfer to Can Tho University's School of Information and Communication Technology. Before field pilots, text processing, fusion, device and network behavior, synchronization and event clustering require validation with rescue professionals. Real-world rescue effectiveness and socioeconomic benefits remain unmeasured.